Homework 01 — Timon Bubnič (63220035)

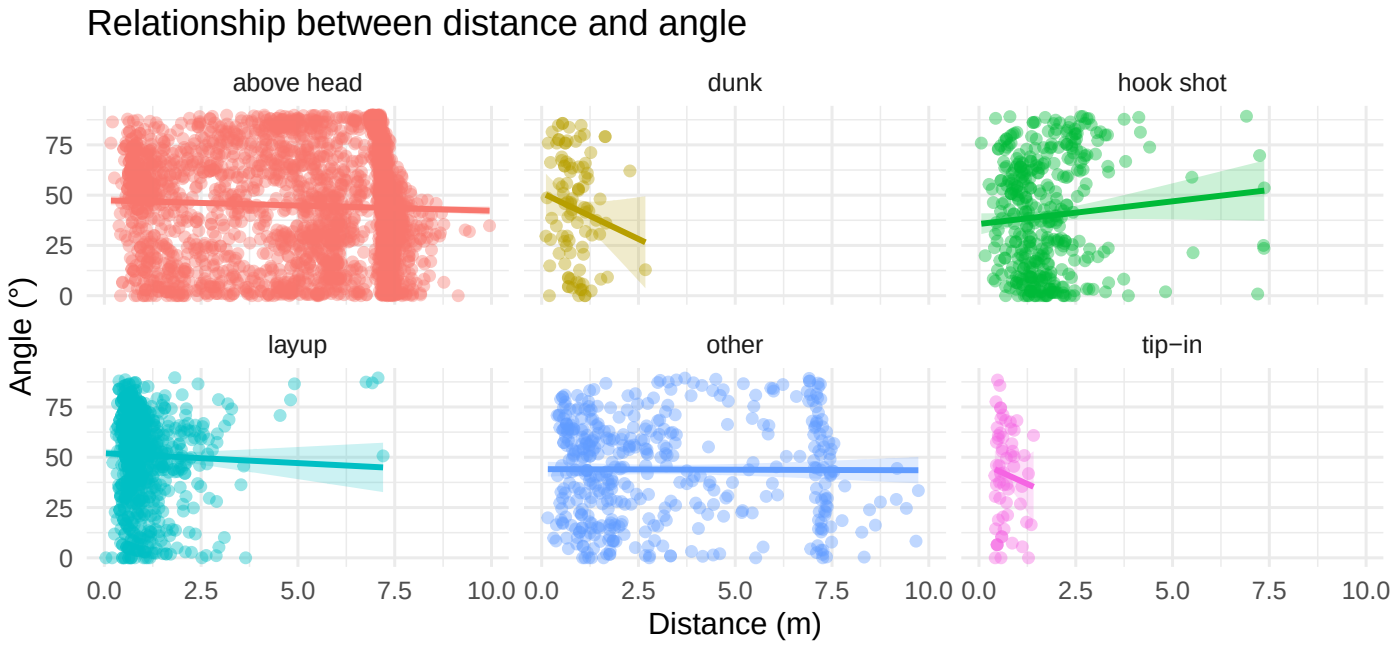
Shot type	Movement	Total shots	Mean distance (m)	Mean angle (°)	SE angle (°)
above head	no	2927	5.513	45.042	0.501
layup	dribble or cut	472	0.984	51.649	0.955
hook shot	no	397	1.731	39.561	1.411
layup	no	391	0.975	49.533	1.108
other	no	221	4.115	45.978	1.695
other	dribble or cut	150	2.264	40.070	1.944
above head	dribble or cut	128	1.633	35.899	2.087
layup	drive	110	1.039	53.700	2.048
dunk	no	99	0.857	43.273	2.671
other	drive	68	2.294	45.685	3.137
tip-in	no	61	0.707	41.445	3.056

Standard error

The sample standard error of the mean angle in each of the grouped shot types and movements is:

$$\mathrm{SE}(\bar{x}) = \frac{s}{\sqrt{n}} = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n(n-1)}}$$

where x_i are the individual angles, $\bar{x}$ their mean, s the sample standard deviation and n the number of shots in the group.



In the figure we can see that across all shot types the angle is scattered over almost the whole range from 0° to 90° at any distance, and the regression lines are nearly flat for above head, layup and other, while for dunks and tip-ins the relationship looks negative (but there are far fewer points, so the confidence intervals are wider) and for hook shots it looks positive. Above head and other shots were taken from 0 to 10 m as its usual for normal jumpshots, while hook shots and layups were mostly concentrated near 0–2.5 m with a few shots from as far as 7.5 m, this again makes sense because of the nature of the shot. Tip-ins and dunks were both taken from within about 2.5 m of the basket which makes sense because these shots are finished right at the rim.